

S M Ferdous

Assistant Professor | Computer Science | UNC Charlotte

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in: <https://www.linkedin.com/in/sm-ferdous/> | 🌐: <http://bit.ly/scholar-smf> | 🐙: <https://github.com/smferdous1>

RESEARCH INTEREST

My research area is at the intersection of **algorithm design and high performance computing**, where I design scalable algorithms for large-scale combinatorial optimization problems. I am particularly interested in leveraging algorithmic tools—including **streaming, randomized, parallel, and distributed approaches**—to address problems across interdisciplinary domains such as **quantum computing, data science, and machine learning**. My work integrates strong theoretical foundations, such as provable approximation guarantees, with a deep emphasis on practical applicability, ensuring that the developed algorithms are both efficient and deployable in real-world settings.

PROFESSIONAL APPOINTMENTS

- **Assistant Professor** Aug '26 – Curr.
Department of Computer Science, CCI
UNC Charlotte
- **Data Scientist** Oct '23 – Aug '26
Data Science and Machine Intelligence Group
Pacific Northwest National Lab, WA, USA.
- **Linus Pauling Distinguished Postdoctoral Fellow** Jun '22 – Oct '23
Pacific Northwest National Lab, WA, USA.
- **Postdoctoral Research Associate** Jan – Jun 2022
Department of Computer Science
Purdue University, USA
- **Graduate Research Assistant** 2015 – 2021
Department of Computer Science
Purdue University, USA
- **PhD Intern** 2017 &
2021 (May – Aug)
Data Science and Machine Intelligence Group
Pacific Northwest National Lab, WA, USA
- **PhD Intern** 2019 (Jun – Aug)
ENSA Group
Nokia Bell Labs, NJ, USA
- **Faculty Member** 2011 – 2015
Department of Computer Science and Engineering
Ahsanullah University of Science and Technology
Dhaka, Bangladesh

EDUCATION

- **PhD in Computer Science**, Purdue University, West Lafayette, Indiana 2015 – 2021
Thesis: “Algorithms for degree-constrained subgraphs and applications”
Advisor: Prof. Alex Pothen | *GPA:* 3.93/4.00.

- **MSc Engg. in Computer Science and Engineering** 2011 – 2014
Bangladesh University of Engineering and Technology (BUET)
Thesis: "Practically Efficient Algorithms for Minimum String Cover and Minimum Common String Partition" | Advisor: Prof. M. Sohel Rahman
- **BSc Engg. in Computer Science and Engineering, BUET** 2006 – 2011
Rank: 9/138, Degree with Honors | GPA: 3.89/4.00.

FELLOWSHIPS

- **Linus Pauling Distinguished Postdoctoral Fellowship** at PNNL. 2022 – 2025
- **John R. Rice Fellowship** for Scientific Computing, Purdue University. 2020 – 2021
- **Ross Fellowship** for incoming graduate student at Purdue University. 2015 – 2016
- **Dean's list and Merit scholarship** at BUET. 2006 – 2011

AWARDS & HONORS

- **Exceptional Contribution Award** at PNNL. 2025
- **Best paper award** at European Symposium on Algorithms (ESA). 2025
- **Best paper award** at ACM International Conference on Supercomputing (ICS). 2025
- **Outstanding Performance Award** at PNNL in recognition of outstanding contribution in exceptional research project leadership, development of high-impact scientific artifacts, and mentorship of multiple interns. 2024
- **Selected for participation** in week-long *Argonne Training Program on Extreme-Scale Computing (ATPESC)* 2021. 2021
- **Travel grant** for SIAM Combinatorial Scientific Computing Workshop in Bergen, Norway. 2018
- **Third best prize** on SIAM Computational Science and Engineering student poster competition at Purdue University. 2017
- **Travel and accommodation grant** for attending week-long SAMSI summer school on optimization at Research Triangle Park, NC. 2016
- **Tenth among 50 teams** in ACM International Collegiate Programming Regional Contest (ICPC) (Dhaka Site). 2008

NEWS COVERAGE

- **PNNL news on our best paper award at ESA25:** <https://www.pnnl.gov/news-media/massive-datasets-meet-their-match>. 2025
- **PNNL news on our IPDPS24 paper:** <https://www.pnnl.gov/news-media/scientists-speed-groundwork-essential-quantum-computing>. Also shared in other coverages, including [Quantum Computing Report](#), [Inside HPC](#), [Quantum Insider](#), and [DOE Office of Science research news update](#). 2025
- **Purdue CS news on Pauling Fellowship:** <https://www.cs.purdue.edu/news/articles/2022/s-m-ferdous-named-pauling-fellow.html>. 2022

PROPOSAL WRITING EXPERIENCE

- **Student writer** of a **\$712,488** *awarded* proposal by DOE ASCR, Title: “Data Summarization and Inference at Scale”, Award number: DE-SC0022260, PI: Alex Pothen. 2021

PUBLICATIONS

The following contains my publications starting from 2015. For the full list, please visit my Google Scholar profile at <http://bit.ly/scholar-smf>. (W) represents workshop papers.

Summary: Over 20 publications, including 12 research papers, in premier (ICORE ranking A/A*) venues such as ESA, SC, IPDPS, KDD, ICS, and JPDC, and one invited paper in the prestigious Acta Numerica journal.

Under Submission/ In Preparation to Submit

Please contact me for the pre-print copies of the following papers.

- [InSub1] Amber Thrall, Vidya Chhabria, **SM Ferdous**, Mahantesh Halappanavar, and Bala Krishnamoorthy. *LockRoute: A Spatial Locking Framework for Parallel Global Routing*. 2025
- [InSub2] Cameron Ibrahim, **SM Ferdous**, Erdal Mutlu, Ilya Safro, and Mahantesh Halappanavar. *Exact Hardware/Software Partitioning for Pathlike Tasks*. 2025
- [InSub3] Baoping Wan, Billy Jin, **SM Ferdous**, and Alex Pothen. *Learning Augmented Minimum Cost Flow*. 2026

Conferences & Workshop

- [KDD26] Siddhartha Shankar Das, Naheed Anjum Arafat, Muftiqur Rahman, **SM Ferdous**, Alex Pothen, and Mahantesh M Halappanavar. *SGS-GNN: A Supervised Graph Sparsification method for Graph Neural Networks*. 2025. arXiv: [2502.10208](https://arxiv.org/abs/2502.10208) [cs.LG]. URL: <https://arxiv.org/abs/2502.10208>
- [ICS26] S M Shovan, Arindam Khanda, **SM Ferdous**, Sajal K. Das, and Mahantesh Halappanavar. “DynLP: Parallel Dynamic Batch Update for Label Propagation in Graph-based Semi-Supervised Learning”. In: *Proceedings of the 40th ACM International Conference on Supercomputing*. ICS '26. Association for Computing Machinery, 2026, pp. 740–751. ISBN: 9798400725227. DOI: [10.1145/3797905.3807875](https://doi.org/10.1145/3797905.3807875). URL: <https://doi.org/10.1145/3797905.3807875>
- [IPDPS(W)26] Michael Mandulak, **SM Ferdous**, Sayan Ghosh, Mahantesh Halappanavar, and George Slota. *ApproxJoin: Approximate Matching for Efficient Verification in Fuzzy Set Similarity Join*. in GrAPL 2026. 2025. arXiv: [2507.18891](https://arxiv.org/abs/2507.18891) [cs.DB]. URL: <https://arxiv.org/abs/2507.18891>
- [HPEC25] Michael Mandulak, Sayan Ghosh, **SM Ferdous**, Mahantesh Halappanavar, and George Slota. “Anonymized Network Sensing using C++ 26 std:: execution on GPUs”. In: *2025 IEEE High Performance Extreme Computing Conference (HPEC)*. IEEE. 2025, pp. 1–7. DOI: [10.1109/HPEC67600.2025.11196472](https://doi.org/10.1109/HPEC67600.2025.11196472)
- [ESA25a] Ahammed Ullah, **SM Ferdous**, and Alex Pothen. “Weighted Matching in a Poly-Streaming Model”. In: *33rd Annual European Symposium on Algorithms (ESA 2025)*. Vol. 351. Leibniz International Proceedings in Informatics (LIPIcs). Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 17:1–17:18. DOI: [10.4230/LIPIcs.ESA.2025.17](https://doi.org/10.4230/LIPIcs.ESA.2025.17). 🏆 **Best Paper Award**.

- [ESA25b] Henrik Reinstädter, **SM Ferdous**, Alex Pothén, Bora Uçar, and Christian Schulz. “Semi-Streaming Algorithms for Hypergraph Matching”. In: *33rd Annual European Symposium on Algorithms (ESA 2025)*. Vol. 351. Leibniz International Proceedings in Informatics (LIPIcs). Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 79:1–79:19. doi: [10.4230/LIPIcs.ESA.2025.79](https://doi.org/10.4230/LIPIcs.ESA.2025.79)
- [ASONAM25] Cameron Ibrahim, **SM Ferdous**, Ilya Safro, Marco Minutoli, and Mahantesh Halappanavar. *ELRUHNA: Elimination Rule-based Hypergraph Alignment*. Accepted at **ASONAM 2025**. 2025. arXiv: [2506.09866](https://arxiv.org/abs/2506.09866) [cs.SI]. URL: <https://arxiv.org/abs/2506.09866>
- [ICS25] Marco Minutoli, Reece Neff, Naw Safrin Sattar, Hao Lu, John Feo, Henning Mortveit, Anil Vullikanti, Dawen Xie, Mandy L Wilson, Gregor von Laszewski, et al. “DIMPLES: Distributed Influence Maximization for Pandemic pLanning on Exascale Systems”. In: *39th ACM International Conference on Supercomputing*. 2025, pp. 718–733. 🏆 **Best Paper Award**.
- [SEA25] Shivaram Gopal, **SM Ferdous**, Alex Pothén, and Hemanta Maji. “GreedyML: A Parallel Algorithm for Maximizing Constrained Submodular Functions”. In: *23rd International Symposium on Experimental Algorithms (SEA 2025)*. Vol. 338. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 19:1–19:21. ISBN: 978-3-95977-375-1. doi: [10.4230/LIPIcs.SEA.2025.19](https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.SEA.2025.19). URL: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.SEA.2025.19>
- [SC24] Michael Mandulak, Sayan Ghosh, **SM Ferdous**, Mahantesh Halappanavar, and George Slota. “Efficient Weighted Graph Matching on GPUs”. In: *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*. IEEE. 2024, pp. 1–16. • Code: 📄 [ECP-ExaGraph/sumac](https://github.com/sumac/ECP-ExaGraph)
- [ESA24] **SM Ferdous**, Bhargav Samineni, Alex Pothén, Mahantesh Halappanavar, and Bala Krishnamoorthy. “Semi-Streaming Algorithms for Weighted k-Disjoint Matchings”. In: *32nd Annual European Symposium on Algorithms (ESA 2024)*. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024, 53:1–53:19. doi: [10.4230/LIPIcs.ESA.2024.53](https://doi.org/10.4230/LIPIcs.ESA.2024.53). • Code: 📄 [smferdous1/GraST](https://github.com/smferdous1/GraST)
- [KDD24] Siddhartha Shankar Das, **SM Ferdous**, Mahantesh M. Halappanavar, Edoardo Serra, and Alex Pothén. “AGS-GNN: Attribute-guided Sampling for Graph Neural Networks”. In: *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. KDD ’24*. ACM, 2024, pp. 538–549. doi: [10.1145/3637528.3671940](https://doi.org/10.1145/3637528.3671940). • Code: 📄 [siddhartha047/AGS-GNN](https://github.com/siddhartha047/AGS-GNN)
- [IPDPS24] **SM Ferdous**, Reece Neff, Bo Peng, Salman Shuvo, Marco Minutoli, Sayak Mukherjee, Karol Kowalski, Michela Becchi, and Mahantesh Halappanavar. “Picasso: Memory-Efficient Graph Coloring Using Palettes With Applications in Quantum Computing”. In: *2024 IEEE International Parallel and Distributed Processing Symposium (IPDPS 2024)*. IEEE. 2024, pp. 241–252. doi: [10.1109/IPDPS57955.2024.00029](https://doi.org/10.1109/IPDPS57955.2024.00029). • Code: 📄 [smferdous1/Picasso](https://github.com/smferdous1/Picasso)
- [SEA24] **SM Ferdous**, Alex Pothén, and Mahantesh Halappanavar. “Streaming Matching and Edge Cover in Practice”. In: *22nd International Symposium on Experimental Algorithms (SEA 2024)*. Schloss Dagstuhl–Leibniz-Zentrum für Informatik. 2024. doi: [10.4230/LIPICS.SEA.2024.12](https://doi.org/10.4230/LIPICS.SEA.2024.12). • Code: 📄 [smferdous1/GraST](https://github.com/smferdous1/GraST)
- [IOS24] Bhargav Samineni, **SM Ferdous**, Mahantesh Halappanavar, and Bala Krishnamoorthy. *Approximate Bipartite b-Matching using Multiplicative Auction*. Accepted as a refereed paper in the 2024 INFORMS Optimization Society conference (**IOS24**). 2024. arXiv: [2403.05781](https://arxiv.org/abs/2403.05781) [cs.DS]. URL: <https://arxiv.org/abs/2403.05781>.

- [SC(W)23] Lizhi Xiang, Arif Khan, **SM Ferdous**, Sr Aravind, and Mahantesh Halappanavar. “CuAlign: Scalable Network Alignment on GPU Accelerators”. In: **SC-W ’23**. Association for Computing Machinery, 2023, pp. 747–755. ISBN: 9798400707858. DOI: [10.1145/3624062.3625129](https://doi.org/10.1145/3624062.3625129). URL: <https://doi.org/10.1145/3624062.3625129>.
- [PDP23] Pasqua D’Ámbra, Fabio Durastante, **SM Ferdous**, Salvatore Filippone, Mahantesh Halappanavar, and Alex Pothén. “AMG Preconditioners based on Parallel Hybrid Coarsening and Multi-objective Graph Matching”. In: *2023 31st Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP 2023)*. 2023, pp. 59–67. DOI: [10.1109/PDP59025.2023.00017](https://doi.org/10.1109/PDP59025.2023.00017).
- [ACDA21] **SM Ferdous**, Alex Pothén, Arif Khan, Ajay Panyala, and Mahantesh Halappanavar. “A parallel approximation algorithm for submodular b -matching”. In: *Proceedings of the First SIAM Conference on Applied and Computational Discrete Algorithms (ACDA 2021)*. SIAM, 2021. DOI: [10.1137/1.9781611976830.5](https://doi.org/10.1137/1.9781611976830.5). • Code: [smferdous1/Submodular-b-matching](https://github.com/smferdous1/Submodular-b-matching)
- [INFOCOM (W)21] Beomyeol Jeon, **SM Ferdous**, Muntasir Raihan Rahman, and Anwar Walid. “Privacy-preserving Decentralized Aggregation for Federated Learning”. In: *2021 IEEE Conference on Computer Communications Workshops, INFOCOM Workshops 2021, Vancouver, BC, Canada, May 10-13, 2021*. First two authors contributed equally. IEEE, 2021, pp. 1–6. DOI: [10.1109/INFOCOMWKSHPS51825.2021.9484437](https://doi.org/10.1109/INFOCOMWKSHPS51825.2021.9484437). • Code: [beomyeol/SecureD-FL](https://github.com/beomyeol/SecureD-FL)
- [SC18] Arif Khan, Krzysztof Choromanski, Alex Pothén, **SM Ferdous**, Mahantesh Halappanavar, and Antonino Tumeo. “Adaptive anonymization of data using b -edge cover”. In: *SC18: International Conference for High Performance Computing, Networking, Storage and Analysis*. IEEE, 2018, pp. 743–753. DOI: [10.5555/3291656.3291735](https://doi.org/10.5555/3291656.3291735).
- [IPDPS18] Arif Khan, Alex Pothén, and **SM Ferdous**. “Parallel algorithms through approximation: b -edge cover”. In: *2018 IEEE International Parallel and Distributed Processing Symposium, IPDPS 2018, Vancouver, BC, Canada, May 21-25, 2018*. IEEE Computer Society, 2018, pp. 22–33. DOI: [10.1109/IPDPS.2018.00013](https://doi.org/10.1109/IPDPS.2018.00013).
- [CSC18] **SM Ferdous**, Alex Pothén, and Arif Khan. “New Approximation Algorithms for Minimum Weighted Edge Cover”. In: *Proceedings of the Eighth SIAM Workshop on Combinatorial Scientific Computing, CSC 2018, Bergen, Norway, June 6-8, 2018*. SIAM, 2018, pp. 97–108. DOI: [10.1137/1.9781611975215.10](https://doi.org/10.1137/1.9781611975215.10).
- [WD16] **SM Ferdous**, Md Mustafizur Rahman, and Mahmuda Naznin. “Finding network connectivity failure in a Wireless Sensor Network”. In: *2016 Wireless Days (WD)*. IEEE, 2016, pp. 1–6. DOI: [10.1109/WD.2016.7461522](https://doi.org/10.1109/WD.2016.7461522).
- [NSysS15] **SM Ferdous** and M Sohel Rahman. “A metaheuristic approach for application partitioning in Mobile System”. In: *2015 International Conference on Networking Systems and Security (NSysS)*. IEEE, 2015. DOI: [10.1109/NSysS.2015.7043520](https://doi.org/10.1109/NSysS.2015.7043520).

Journals

- [J5] Reet Barik, Wade Cappa, **SM Ferdous**, Marco Minutoli, Mahantesh Halappanavar, and Ananth Kalyanaraman. “GreediRIS: Scalable influence maximization using distributed streaming maximum cover”. In: *Journal of Parallel and Distributed Computing* 198 (2025), p. 105037. ISSN: 0743-7315. DOI: <https://doi.org/10.1016/j.jpdc.2025.105037>. URL: <https://www.sciencedirect.com/science/article/pii/S0743731525000048>

- [J4] Seher Acer, Ariful Azad, Erik G. Boman, Aydın Buluç, Karen D. Devine, **SM Ferdous**, Nitin Gawande, Sayan Ghosh, Mahantesh Halppanavar, Ananth Kalyanaraman, Arif Khan, Marco Minutoli, Alex Pothén, Sivasankaran Rajamanickam, Oguz Selvitopi, Nathan R. Tallent, and Antonino Tumeo. “EXAGRAPH: Graph and Combinatorial Methods for Enabling Exascale Applications”. In: *The International Journal of High Performance Computing Applications* (2021), p. 10943420211029299. doi: [10.1177/10943420211029299](https://doi.org/10.1177/10943420211029299).
- [J3] Alex Pothén, **SM Ferdous**, and Fredrik Manne. “Approximation algorithms in combinatorial scientific computing”. In: *Acta Numerica* 28 (2019), pp. 541–633. doi: [10.1017/S0962492919000035](https://doi.org/10.1017/S0962492919000035).
- [J2] **SM Ferdous** and M. Sohel Rahman. “Solving the Minimum Common String Partition Problem with the Help of Ants”. In: *Math. Comput. Sci.* 11.2 (2017), pp. 233–249. doi: [10.1007/s11786-017-0293-5](https://doi.org/10.1007/s11786-017-0293-5).
- [J1] **SM Ferdous** and M. Sohel Rahman. “An Integer Programming Formulation of the Minimum Common String Partition Problem”. In: *PLOS ONE* 10.7 (July 2015), pp. 1–16. doi: [10.1371/journal.pone.0130266](https://doi.org/10.1371/journal.pone.0130266).

Selected Talks

- 2025 **SM Ferdous**. “Large-scale Graph Analytics: A case for memory efficiency”. Presented in GraphBlas BOF, SC 2025, St. Louis, MO. 2025.
- 2024 **SM Ferdous**. “Semi-streaming algorithms for weighted k-Disjoint matching”. Presented in SIAM Discrete Math 2024, Spokane, WA. 2024.
- 2021 **SM Ferdous** and Alex Pothén. “Locality Matters! Efficient algorithms for submodular b -matching”. Presented in SIAM CSE 2021, Happened virtually. 2021.
- 2019 **SM Ferdous** and Alex Pothén. “Efficient Algorithms for Degree Constrained Subgraphs with Applications”. Presented in SIAM CSE 2019, Spokane, WA. 2019.
- 2018 **SM Ferdous** and Alex Pothén. “New Approximation Algorithms for Minimum Weight Edge Cover”. Presented in SIAM AN 2018, Portland, OR. 2018

TEACHING EXPERIENCE

- **Ahsanullah University of Science & Technology** Mar – Aug 2015
Assistant Professor, Dept. of Computer Science & Engineering
Courses Taught: Elementary Structured Programming (C)
- **Ahsanullah University of Science & Technology** Oct '11 – Feb '15
Lecturer, Dept. of Computer Science & Engineering
Courses Taught: Mathematical Analysis of Computer Science, Database, Operating System, Pattern Recognition

RESEARCH MENTORING

- **Reece Neff**, PhD student at NCSU 2023 – 2025
Research: Memory-efficient parallel algorithms.
Co-author of [IPDPS24, ICS25]
- **Siddhartha Shankar Das**, PostDoc, PNNL 2023 – 2025
Research: Sparsification algorithms for GNN.
Co-author of AGS-GNN [KDD24, KDD26]

- **Bhargav Samineni**, Capital One 2022 – 2024
Research: Semi-streaming graph algorithms.
Co-author of k -DM [ESA24, IOS24]
- **Michael Mandulak**, PhD student at RPI 2022 – 2026
Research: Parallel matching algorithms and applications.
Co-author of [SC24, IPDPS(W)26]
- **Reet Barik**, PostDoc, ANL. 2022 – 2024
Research: Efficient Distributed Influence Maximization
Co-author of GreediRIS [J5]
- **Shivaram Gopal**, PhD student at Purdue. 2021 – Curr.
Research: Efficient Distributed Submodular Optimization
Co-author of GreedyML [SEA25]
- **Cameron Ibrahim**, PhD student at Univ. of Delaware. 2023 – Curr.
Research: Hypergraph Alignment
Co-author of ELRUHNA [ASONAM25]
- **S M Shovan**, PhD student at Missouri S&T. 2024 – Curr.
Research: Parallel Dynamic Algorithms
Co-author of Dyn-LP [ICS26]

PROFESSIONAL SERVICES & MISCELLANY

- Journal review:
 - VLDB Journal
 - Parallel Computing
 - SIAM Journal on Matrix Analysis and Applications (SIMAX)
 - ACM Transactions on Parallel Computing (TOPC)
 - IEEE Transactions on Parallel and Distributed Systems (TPDPS)
 - Springer Journal of Combinatorial Optimization (JOCO)
 - ACM Journal of Experimental Algorithmics (JEA)
 - PLOS ONE
- Conference and Workshop committee member:
 - Demo Track, WSDM 2026
 - Algorithm Track: SC 2025, SC 2026
 - European Symposium on Algorithms (ESA - Track B) 2025
 - SIAM Symposium on Algorithm Engineering and Experiments (ALENEX) 2026 2027
 - IEEE International Parallel & Distributed Processing Symposium (IPDPS) 2024, 2025
 - SIAM Conference on Applied and Computational Discrete Algorithms (ACDA) 2025
 - Workshop on Graphs, Architectures, Programming, and Learning (GrAPL) 2025 2026
- Conference review (External):
 - SIAM Conference on Applied and Computational Discrete Algorithms (ACDA) 2021, 2023
 - Symposium on Experimental Algorithms (SEA) 2022
 - European Symposium on Algorithms (ESA) 2024
 - International Conference and Workshops on Algorithms and Computation (WALCOM) 2019
- Community Services
 - General Secretary (2019-2020), Bangladeshi Student Association (BDSA), Purdue.
 - Webmaster (2015-2016, 2017-2018), Bangladeshi Student Association (BDSA), Purdue.